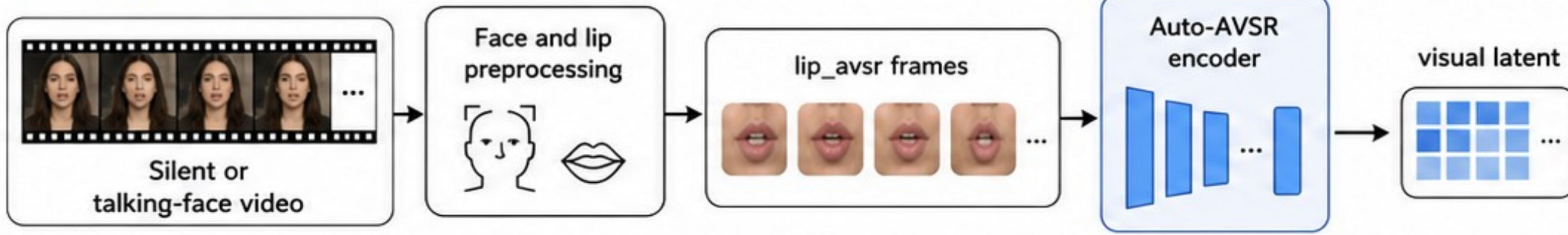


Audio-Centric Visual Speech Reconstruction

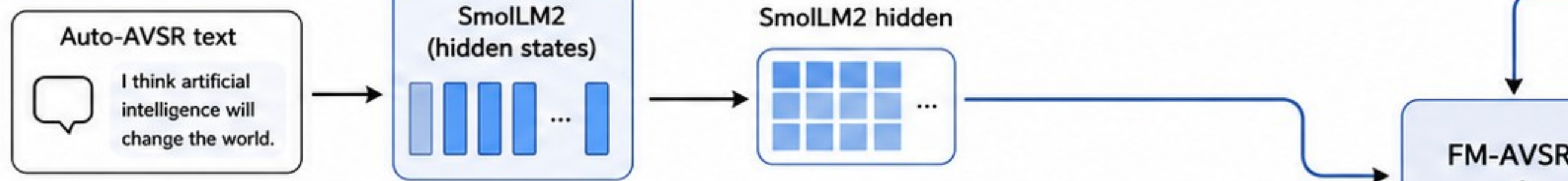
Changxun Pan, Zihan Guo, Anqiao Fu

Generate speech by reconstructing Mimi audio latents, not by cascading lip $\rightarrow$ text $\rightarrow$ TTS

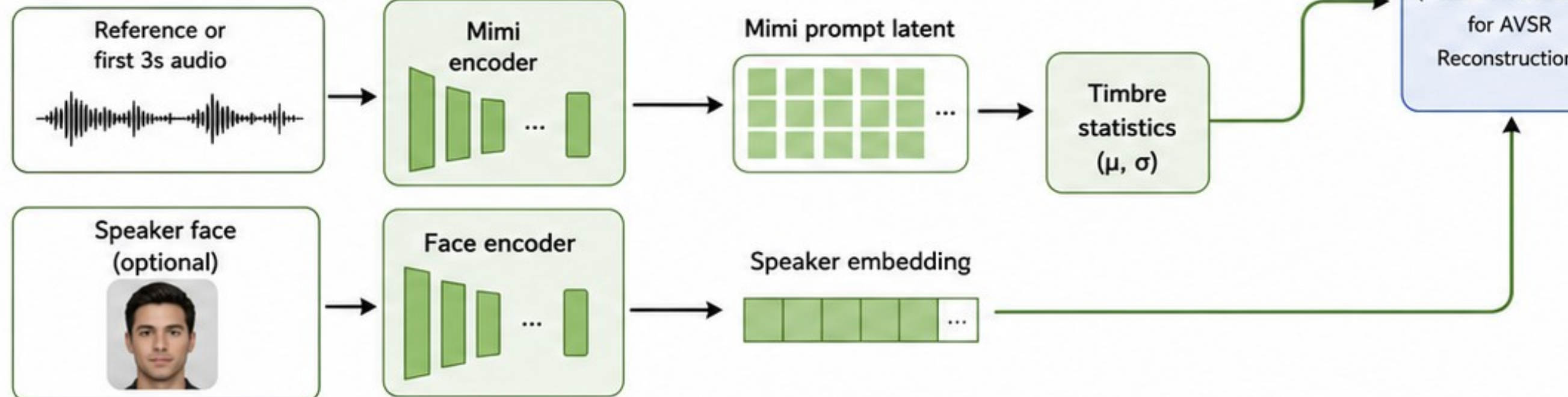
1) Visual (lip-reading) condition



2) Text (semantic) condition



3) Speaker (timbre) condition



1 CORE ARCHITECTURE

- Conditions: lip/video | Weak text hidden states | Speaker embedding | 3.04s Mimi audio prompt | Prompt timbre stats
- Target:** normalized Mimi latents $\rightarrow$ waveform via Mimi decoder (offline, 24 kHz mono)
- Key:** text is NOT the main generation path — lip and audio-prompt latents carry most of the signal

2 PROJECT OVERVIEW

- Main Claim**
Do not turn visual speech generation into a strict lip $\rightarrow$ text $\rightarrow$ TTS cascade. Text is a weak semantic/alignment cue.
- Route**
Lip motion, weak LM text, speaker identity, and reference-timbre prompt are fused to reconstruct continuous Mimi audio latents.
- Evidence**
Trained on 59,144 LRS3 lip-AVSR clips; final checkpoint reaches corr 0.5818 / MSE 0.6676 / MAE 0.6018.

3 DATA + TRAINING SIGNALS

59,144 lip-AVSR clips | fixed 1,000 val

- lip_avsr.npy — Auto-AVSR lip crop
- latentavsr_enc_lipavsr.npy — AVSR visual encoder state
- avsr_enc_lipavsr.npy — AVSR visual encoder state
- avsr_text_lipavsr.txt — noisy text, weak semantic aid
- SmolLM2 hidden — aligned text-token hidden states
- speaker/timbre — speaker_emb + Mimi prompt stats
- target — Mimi audio latent to reconstruct
- lip crops + weak text + timbre prompt

4 TEXT ABLATION

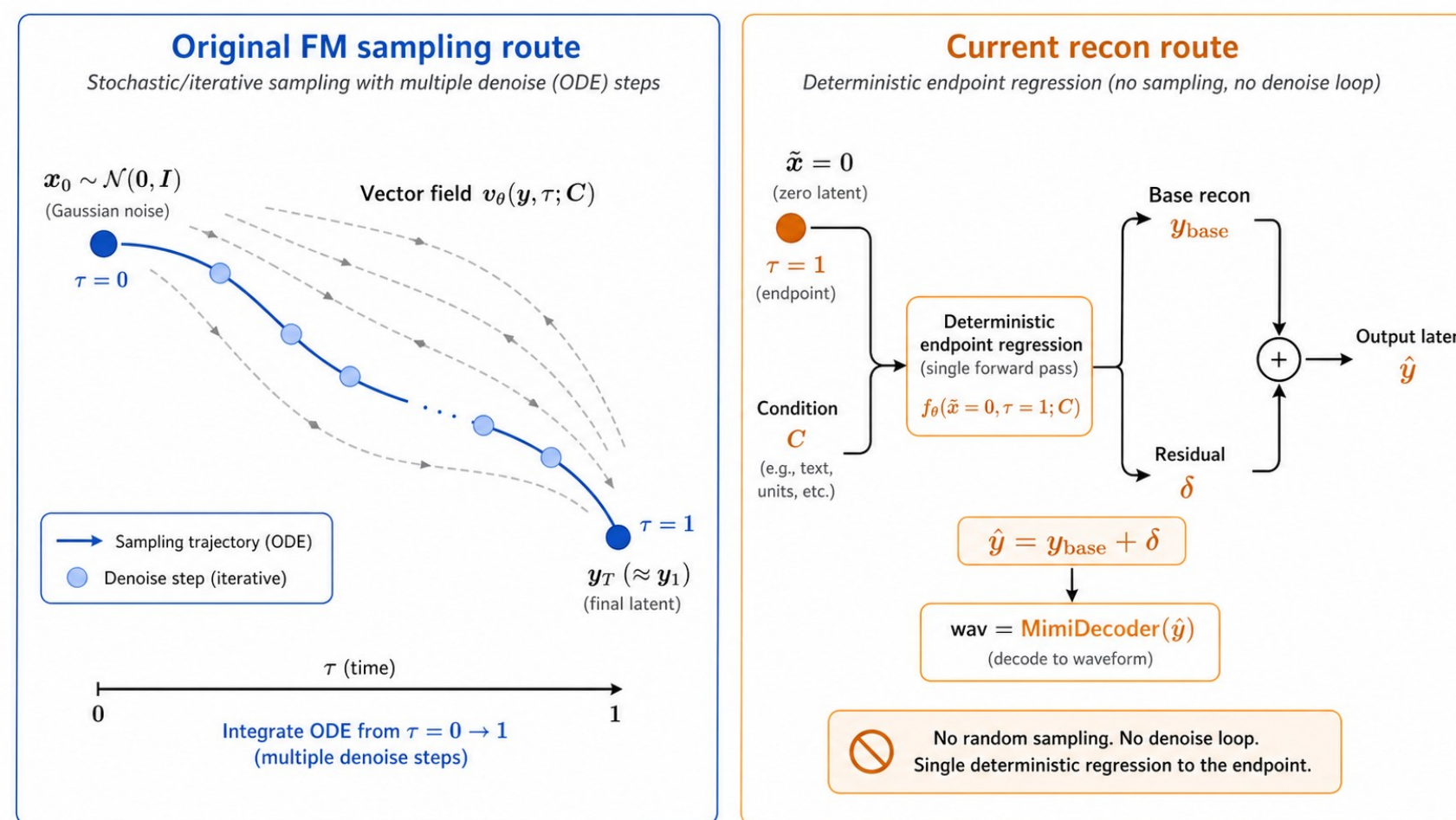
GT transcript is not the core bottleneck.

Text source	Corr	MSE	MAE
GT text + timestamps	0.5818	0.6676	0.6018
AVSR text + timestamps	0.5783	0.6716	0.6037
AVSR text + uniform	0.2868	1.684	1.024
V5 text + uniform	0.2861	1.685	1.024

- AVSR vs GT: $\Delta\text{corr} = 0.0035$ only.
- V5 (WER 29%) vs AVSR (WER 20%): $\Delta\text{corr} = 0.0007$.
- Alignment quality (timestamps vs uniform) = $\Delta 0.29$.**

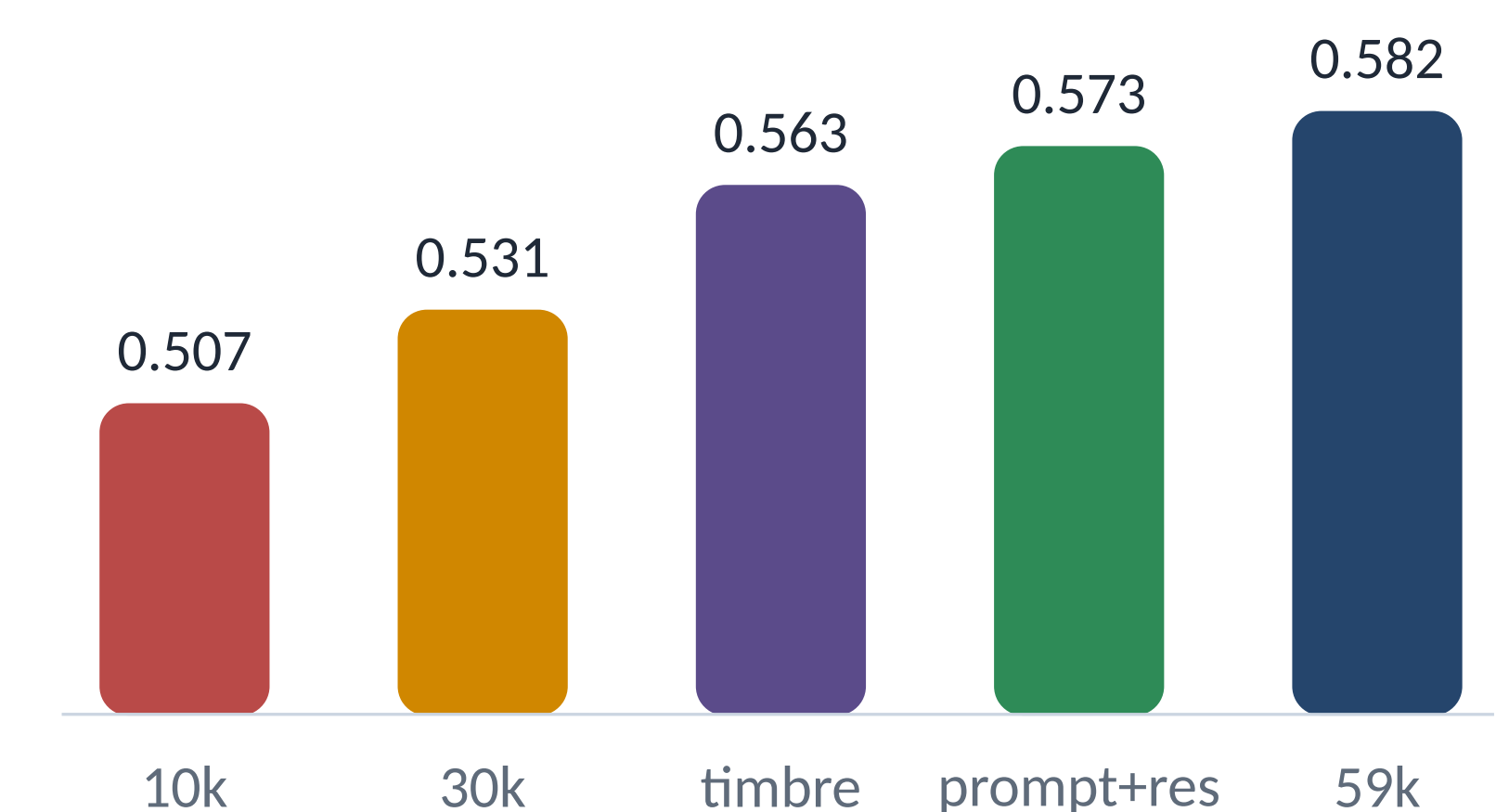
alignment > accuracy

5 RESIDUAL RECON



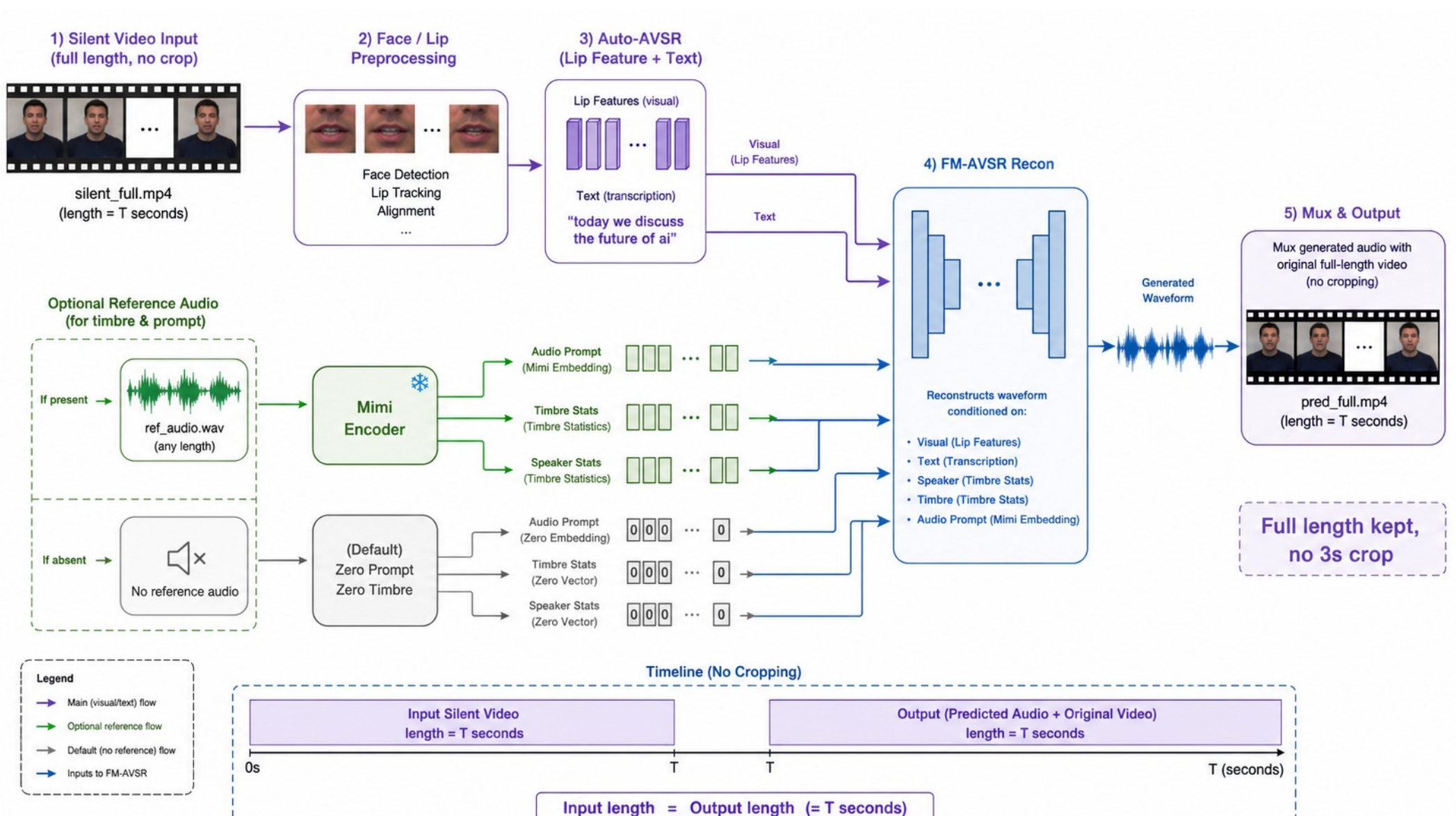
- FM sampling (NFE=10) stalls at corr ≈ 0.35 .**
Fix: set $\tilde{x}=0, \tau=1$ — predict endpoint directly.
- Frozen base reconstructor predicts latent.
 - Trainable residual head corrects errors. Doubles corr vs naive endpoint.
- $\hat{y} = f_{\text{base}}(C) + \Delta(C)$

6 EXPERIMENTAL COMPARISON

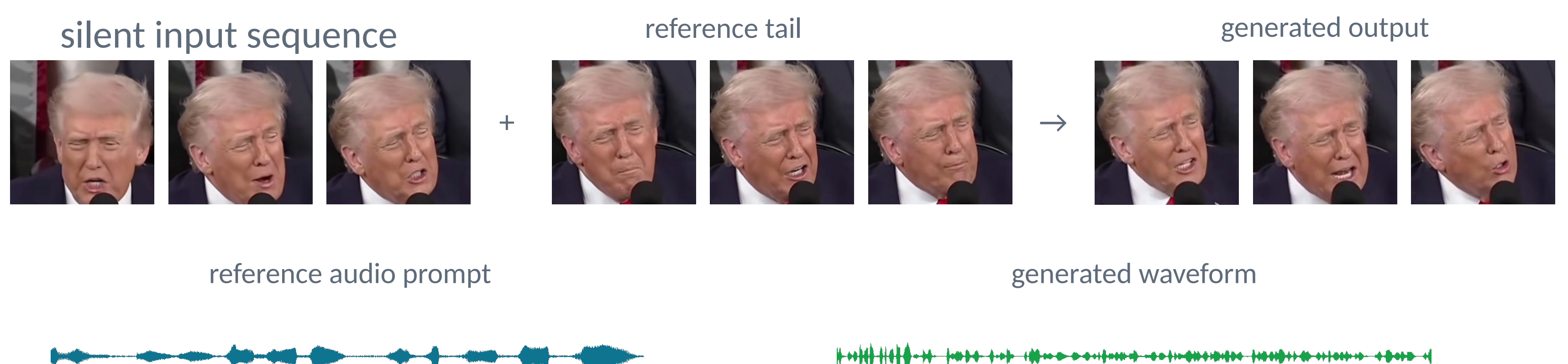


- Scale + timbre + residual progression (corr)
- 10k visual prior: 0.507.
 - Final 59k prompt-stats residual: 0.582.
 - Timbre/audio-prompt conditioning and residual recon produce the larger gains.
- Final corr 0.5818 | MSE 0.6676 | MAE 0.6018

7 RAW / SILENT VIDEO PIPELINE



8 SILENT $\rightarrow$ VOICED DEMO



Silent video + reference audio $\rightarrow$ AVSR text $\rightarrow$ Mimi recon $\rightarrow$ Voiced MP4
Trump corr: 0.3035 $\rightarrow$ 0.3869 with final lip-AVSR preprocessing path.

9 STREAMLIP V5: SELF-TRAINED VSR BRANCH

Architecture



lr groups: OLMo pretrained 1e-5 | proj+CA 3e-4 | gates 1e-4 | OLMo pre-finetuned on LRS3

Beam sweep on long sentences ($\geq 10w, \geq 4s, 126$ clips)

beam	AVSR WER	V5 WER	gan
1 greedy	11.1%	21.9%	10.8nn
2	10.3%	16.2%	5.9nn
3 ✓ best	10.2%	15.4%	5.2nn
4	10.3%	15.6%	5.3nn
40	9.9%	15.9%	6.0pp

10 TAKEAWAY

11 REPRODUCE / ARTIFACTS